
AI-Assisted Penetration Testing

Continuous security assessment of APIs, endpoints, domains, and network zones

Using the **AccuKnox AgentZ** Zero Trust agent platform

Attack surfaces are no longer static. APIs ship weekly, endpoints spin up and down, domains multiply, and cloud network zones shift by the hour. A point-in-time pen test, run once a quarter by scarce human specialists, cannot keep pace. **AccuKnox AI-Assisted Penetration Testing** replaces the periodic engagement with continuous, autonomous adversarial assessment across your APIs, applications, endpoints, domains, and network zones, with every testing action governed inside a sandbox you control.

24/7

Continuous, autonomous
assessment

0

Standing agent privileges

100%

Actions logged and
policy-governed

1

Unified platform for the
whole estate

THE CHALLENGE

Why traditional pen testing breaks down

Security teams are asked to defend an estate that changes faster than they can test it by hand. Three gaps compound.

Coverage gaps

A manual test samples a slice of the estate at a single point in time. New API endpoints, subdomains, and cloud network zones provisioned the next day go untested until the next engagement, often months later.

Speed and cost

Skilled human testers are scarce and expensive. Backlogs mean critical exposures sit undiscovered while attackers scan continuously.

APIs move fastest of all

Every release adds routes, parameters, and versions. Undocumented and shadow APIs never make it into a test scope, so broken authorization, broken authentication, and overexposed data stay live until someone finds them.

The result is a widening window between when an exposure appears and when anyone finds it. Closing that window takes assessment that is continuous, automated, and safe to run against production-adjacent assets.

THE SOLUTION

Always-on assessment, using AgentZ

AccuKnox runs penetration testing as an always-on capability. Autonomous testing agents run using **AgentZ**, the AccuKnox Zero Trust sandbox platform for building, running, and governing production agents. AgentZ gives each agent the skills, tool access, and workflows it needs to probe a target, while enforcing isolation, ephemeral credentials, and policy mediation so the assessment can never become the incident.

Asset domain	Continuous assessment coverage
APIs	Discovery of documented, undocumented, and shadow endpoints, then authenticated and unauthenticated probing for broken object and function level authorization, broken authentication, injection, and data overexposure.
Web applications	Crawl and active testing of public and logged-in surfaces, in baseline or full mode, against injection, cross-site scripting, misconfigured headers, and session handling flaws.
Domains	Subdomain and attack-surface discovery plus exposed-service enumeration across the external footprint.
Endpoints	Workstations, servers, and containers tested for exploitable vulnerabilities, misconfigurations, weak credentials, and exposed services as they come online.
Network zones	Segmentation and lateral-movement testing across cloud and on-prem zones, to confirm trust boundaries hold under real attack paths.
Cloud and Kubernetes	Entitlement, workload, and configuration exposure mapped and exercised against real exploit paths.

HOW IT WORKS

The assessment loop

Testing runs as a closed, repeating loop. Every stage is executed by testing agents operating inside the AgentZ sandbox, so autonomy never comes at the cost of control.

01 Discover

Agents continuously map the live attack surface: API endpoints, applications, subdomains, exposed services, network zones, and cloud entitlements. Nothing is tested against a stale inventory.

02 Prioritize

Findings are scored by exploitability and business exposure, aligning with **Continuous Threat Exposure Management**, so teams fix what attackers would actually reach first.

03 Simulate and exploit

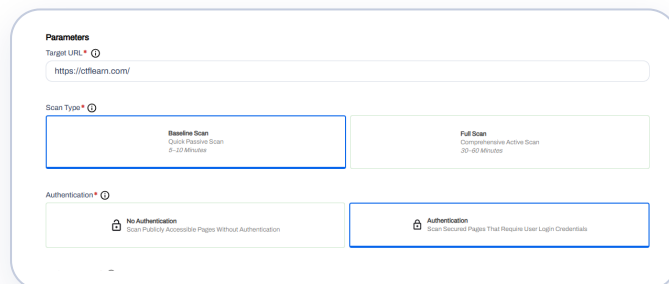
Testing agents safely reproduce real adversary technique, from service exploitation and authorization abuse to lateral movement, inside controlled and isolated conditions.

04 Validate and report

Confirmed exposures are documented with evidence, location, and reproduction steps, which cuts false positives and gives remediation teams prioritized results they can act on.

05 Re-test

After a fix, agents automatically re-run the relevant tests on a schedule to confirm closure, turning pen testing from a report into a continuously verified control.



Parameters

Target URL 

https://cfllearn.com/

Scan Type 

Baseline Scan
Quick Recurrent Scan
5-10 Minutes

Full Scan
Comprehensive Active Scan
20-60 Minutes

Authentication 

No Authentication
Scan Publicly Accessible Pages Without Authentication

Authentication
Scan Secured Pages That Require User Login Credentials

Scope a target: URL, scan depth, authentication.

☐	Last Seen	Vulnerability Name	Asset Name	Description	Risk Factor	Location	
☐	2025-11-14 11:04:00	Re-examine Cache-control Dir	https://accuknox.com	The cache-control header has	Informational	https://accuknox.com/	
☐	2025-11-14 11:04:00	Insufficient Site Isolation Again	https://accuknox.com	Cross-Origin-Embedder-Policy	Low	https://accuknox.com/	
☐	2025-11-14 11:04:00	Sub-Resource Integrity Attribut	https://accuknox.com	The integrity attribute is missin	Medium	https://accuknox.com/	
☐	2025-12-05 15:38:54	Content Security Policy (CSP)	https://www.accuknox.com	Content Security Policy (CSP)	Medium	GET, https://www.accuknox.co	
☐	2025-11-14 11:04:00	Content Security Policy (CSP)	https://accuknox.com	Content Security Policy (CSP)	Medium	https://accuknox.com/	
☐	2025-11-14 11:04:00	Timestamp Disclosure - Unix	https://accuknox.com	A timestamp was disclosed by	Low	https://accuknox.com/	
☐	2025-11-14 11:04:00	Information Disclosure - Supple	https://accuknox.com	The response appears to cont	Informational	https://accuknox.com/	
☐	2025-11-14 11:04:00	Modern Web Application	https://accuknox.com	The application appears to be	Informational	https://accuknox.com/	

Findings carry risk factor, location, and the fix.

BUILT ON ZERO TRUST

Why autonomous testing stays safe to run

Handing an autonomous agent the ability to run exploits is only responsible if that agent is contained. AgentZ enforces containment by design.

Sandbox isolation

Agents run with process, filesystem, credential, and network isolation enforced at the kernel by KubeArmor, eBPF, and Linux Security Modules, not trusted to the application layer.

Zero standing privilege

Agents receive ephemeral, just-in-time credentials scoped to a single task, so a compromised or misbehaving agent has nothing durable to abuse.

Policy mediation

Real-time policy validation inspects every agent action and tool call before it leaves the sandbox, blocking unsafe execution and data-leak paths.

Full auditability

Every agent action is logged and governed, producing the evidence trail a compliance review asks for.

BUSINESS OUTCOMES

What security teams gain

Close the exposure window

Continuous testing surfaces new weaknesses as assets appear, not months later at the next engagement.

Test every API, not a sample

Discovery keeps the scope current, so new and undocumented endpoints get probed in the same pass as the ones you already knew about.

Scale scarce expertise

Autonomous agents handle breadth and repetition, freeing human experts for the highest-value creative testing.

Prioritize what matters

CTEM-aligned scoring focuses remediation on the exposures an attacker could actually reach and exploit.

Prove control, not just effort

Automatic re-testing verifies fixes, converting pen testing into a continuously validated security control.

Stay audit-ready

Governed, logged agent activity gives auditors a dated, reproducible record of what was tested and what was fixed.

WHERE IT FITS

One capability in the AccuKnox platform

AI-assisted pen testing works alongside the rest of the Zero Trust platform, unifying assessment, protection, and governance across API, application, cloud, and endpoint workloads.

API Security

ASPM (SAST, DAST, SCA)

CTEM

CSPM

CWPP

KSPM

Compliance

See continuous pen testing against your own estate

Book a demo and we will map your APIs, applications, endpoints, domains, and network zones to a live assessment loop, then walk the findings, the evidence, and the re-test schedule with your team.

Book a demo

API Security

Docs: unauthenticated scan

Docs: authenticated scan

Docs: API security